

Research Paper 07 Summary

Daytime Multisensor Satellite Analysis of Industrial Thermal Emission Sources

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Faruolo et al. (2021), *Remote Sensing*

1 Basic Information

- **Paper Title:** A Daytime Multisensor Satellite Analysis of Industrial Thermal Emission Sources
- **Authors:** Mariapia Faruolo, Irina Coviello, Alfredo Falconieri, Teodosio Lacava, and Nicola Pergola (2021)
- **Sensors Covered:** High-resolution daytime imagery from Landsat-8 (30 m) and Sentinel-2 (10 m–20 m).

2 What is this Paper About? (In Simple Words)

Most gas flaring studies only look at nighttime satellite images because darkness makes heat easy to spot. However, daytime fire alerts are much more common and much harder to classify because the sun heats up roads, bare fields, and metal rooftops.

- **The Daytime Solution:** The authors developed a method to detect and analyze industrial heat sources during broad daylight across major European factories (steel mills, oil refineries, cement plants, and power stations).
- **Normalized Hotspot Indices (NHI):** Instead of just looking at raw temperature, the method calculates mathematical ratios comparing Shortwave Infrared (SWIR) and Near-Infrared (NIR) light channels.
- **Why NHI Works in Daylight:** Solar radiation reflects strongly across all visible bands, but extreme industrial heat produces a unique spike in SWIR wavelengths that sunlight alone cannot create.

3 Key Concept: Separating Industrial Types During the Day

The paper provides a spectral breakdown for different heavy industrial categories:

- **Oil Refineries & Petrochemical Plants:** Characterized by tall, concentrated flare stacks with extreme SWIR radiance.
- **Steel Works & Blast Furnaces:** Show large, high-heat signatures near slag pits and molten metal processing units.

- **Cement Kilns:** Produce moderately high, continuous thermal radiance from rotary kilns.
- **Hot Bare Soil / Rooftops:** High surface temperature, but zero spectral spike in SWIR combustion channels.

4 How This Helps Our Project (PS26162)

4.1 1. Solving the Daytime False Alarm Problem

NASA FIRMS sends fire alerts during the day (morning and afternoon satellite overpasses). This paper provides the exact spectral index formulas (Normalized Hotspot Indices) to prevent hot soil, concrete, or factory rooftops from being falsely flagged as industrial fires.

4.2 2. High-Resolution Verification Layer (Sentinel-2 / Landsat-8)

In our system architecture, when a coarse 375 m FIRMS alert is detected over an industrial area:

- Our backend can automatically pull high-resolution (10 m–20 m) Sentinel-2 imagery.
- Using the NHI formulas from this paper, the system can zoom in and pinpoint the exact flare stack or furnace unit within a complex factory boundary.

4.3 3. Fine-Grained Industrial Sub-Classification

Instead of just classifying an alert generically as “Industrial”, the spectral patterns in this paper help our AI model sub-classify the detection into specific industries:

Classes: [Refinery / Flare, Steel Mill / Smelter, Cement Kiln, Wildfire, Agricultural Burn]
(1)

4.4 4. Published Precision/Recall Benchmarks

The paper provides a detailed confusion matrix for daytime industrial classification, giving our team documented accuracy benchmarks to cite and compare against.